passing across the 2 to 7 km/h range on three meshes, which is a selection-time record for a different checkpoint than the one described above. An earlier six-mesh evaluation of the rebuilt pipeline reports seven gates passing and *ground contact failing*, at 0.6486 against the 0.700 threshold and a margin of −0.0514, which was at the time the one genuine performance gap in the stack. Neither record certifies the checkpoint this chapter selects, and neither is the evidence used in Chapter 7: the closed-loop comparison there re-scores a single checkpoint per stack under one protocol, at one speed, over the full path, on the same meshes and the same eleven families. The earlier ground-contact failure is still the right expectation to carry into that chapter. It is measured on a worst-mesh, worst-family aggregation, which is the statistic that the pooled per-cell means of Table 7.3 do not report, and it identifies ground contact on the ROM plant as the gate most likely to bind, a prediction Chapter 7 is able to check rather than assume.

6.6 IF: Non-Adaptive Learned Baseline

IF subclasses the R task and changes exactly one thing. The policy still emits desired cylinder motion, but that command is now realised by the frozen inverse model of Section 5.2, which converts it to a valve command, followed by the frozen strict forward model of Section 5.3, which converts the valve command to a predicted extension increment. PhysX receives the same bounded cylinder target as before. Task, terrain families, meshes, observation contract, action chain, reward, gates and the PPO update are inherited unchanged, so the actuator representation is the only free variable between the two stacks.

The stack matters to the argument twice over. It is the honest control for R, whose actuator coincides with the deployment plant, and it is the stack in which the inverse model is carried in the command path and is adequate there. It is not the reference against which AF is measured: the two do not share the action contract, and the adaptive claim is made against that stack's own frozen base instead (Section 6.9.2).

Robust is used here in the sense of Section 2.6: the stack was trained across the randomised plant family and applies one fixed response to whichever member of that family it meets, performing no online estimation of plant characteristics and conditioning on no plant identity. It does not mean the policy is blind to plant variation. The twelve model-error channels of Section 6.2 carry the actuator's standing residual, and that residual is itself plant-dependent: under a given plant the actuator misses its commanded increment by a characteristic amount, and the policy acts on that miss. What the stack lacks is not information about the consequences of plant variation but any representation of *which* plant produced them.

Two properties of the composed path shape its behaviour, and both were established offline. The valve command is accumulated rather than issued absolutely and is clipped to the per-axis limits $[0.95, 0.95, 0.80, 0.80]$, so the actuator path carries its own saturation boundary independently of the action chain. And only the predicted extension increment is integrated, since the forward model's velocity output failed validation on two axes (Section 5.3) and is carried as a diagnostic. This stack's standing model error is therefore visible to the policy through the twelve model-error channels of Section 6.2, which is what makes the shared observation contract fair across actuators rather than merely identical.

The promoted policy's run manifest records all eight gates passing across the 2 to 7 km/h speed range on three held-out meshes. That is the selection-time evidence on which the